MODULE 3

Shock and Vibration sensor

A shock and vibration sensor is a device that detects, measures, and analyses mechanical oscillations, sudden impacts, or other physical movements in a system. These sensors convert a mechanical motion into an electrical signal that can be processed and interpreted to monitor the health of equipment, ensure safety, and improve efficiency. vibration is an oscillatory movement of small amplitude. All bodies have a vibration signal in which are reflected each of their characteristics.

A vibration sensor is a device that measures the amount and frequency of vibration in a given system, machine, or piece of equipment. Vibration sensors can be used to give maintenance teams insight into conditions within key assets that might lead to equipment failure, allowing them to predict the maintenance of the machinery, to reduce overall costs and increase the performance of the machinery.

How shock and vibration sensors work

Different types of sensors use varying principles to operate, but the general process involves converting a physical motion into a measurable electrical signal.

- **Transduction:** A sensing element, or transducer, is at the core of the sensor. For example, in a piezoelectric sensor, a crystal is placed under stress when vibrations or impacts occur. The crystal then produces a voltage proportional to the force applied.
- **Signal processing:** The electrical signal generated by the transducer is then processed. Modern sensors often include a built-in analog-to-digital converter (ADC) and electronics to condition and filter the signal, ensuring it is clean and accurate.
- **Output:** The processed signal is then sent to an external interface. Depending on the sensor type, the output can be either an analog voltage or a digital value. This output is analysed to understand the characteristics of the vibration, such as its frequency, amplitude, and acceleration.

Types of shock and vibration sensors

1. **Accelerometers**

- **Piezoelectric accelerometers:** These are highly durable, wide-range, and accurate. They use a piezoelectric crystal that produces a charge when exposed to mechanical strain from acceleration.
- **Capacitive MEMS accelerometers:** This type uses micro-electro-mechanical systems (MEMS) with a proof mass and fixed fingers that change capacitance when moved by vibration. They are compact, low-power, and often used in wireless sensors.
- **Piezoresistive accelerometers:** Similar to piezoelectric types, these use a piezoresistive material, such as silicon, that changes its electrical resistance when under mechanical strain.

2. **Other types of sensors**

- **Velocity sensors:** Also known as velocity transducers, these measure the velocity of a vibrating surface and are particularly useful for low-frequency applications, like seismic monitoring.
- **Displacement sensors:** These are non-contact sensors that measure the relative motion between an object and a reference point. They are well-suited for monitoring shaft movement in sleeve-bearing machines.
- **Strain gauges:** These sensors consist of a foil with a conductive grid. As a material vibrates and deforms, the resistance of the grid changes, which is then measured.
- **Shock sensors (spring-type):** These simple switches often use a spring-loaded conductive mass. When a sudden impact or shock occurs, the mass makes contact with the leads, changing the electrical signal.

Common applications

Shock and vibration sensors are crucial in many industries for monitoring asset health, ensuring safety, and enabling predictive maintenance.

- **Industrial machinery:** Monitoring for early signs of failure, such as imbalances, loose parts, or bearing defects in equipment like pumps, motors, and gearboxes.
- **Predictive maintenance:** Tracking vibration data over time to predict when equipment might fail. This allows for scheduled maintenance, which reduces unplanned downtime and repair costs.
- **Aerospace:** Monitoring vibrations in aircraft and rockets during flight, takeoff, and landing to ensure structural integrity and flight safety.
- **Automotive:** Detecting sudden impacts for airbag deployment, monitoring component fatigue, and improving vehicle reliability.
- **Construction:** Assessing ground stability and monitoring the structural health of buildings, bridges, and dams. Building services is another massive area where the performance of motors, pumps and fans is critical in public buildings such as hospitals, airports, railway stations, office blocks, hotels, etc.
- **Security systems:** Detecting forced entry or tampering by sensing impacts and vibrations on doors, windows, and walls.
- Pulp and Paper
- Food and Beverage
- Oil and Gas, Refining, Petrochemicals
- Power Generation (fossil fuel, nuclear, hydro, wind turbines)
- Machine tools
- Metalworking
- Mining, Mineral Processing, and Cement

The most common industrial applications of **vibration sensors** include rotating equipment such as pumps, motors, fans, compressors, gearboxes, gas turbines, bearings on conveyor systems, wind turbine gearboxes, wind turbine generators, dryer sections (pulp & paper), ovens & kilns (mineral processing), and hot rolling mills (metalworking).

The **water industry** now monitors its many water processing pumps from those situated in remote pumping stations to those in large scale sophisticated water treatment and de-salination sites.

The **food processing industry** needs to monitors its many rotating machinery parts in vibratory conveyor systems, cooling fans, food pumps and mixers, the chip, can cause cost losses of many thousands of dollars.

Vibration sensors play a vital role in the **oil and gas industry** by ensuring that the production machinery is operational and the downtime is minimized, thereby leading to a reduction in operating costs and an increase in productivity and monetary gains. These sensors help reduce unplanned outages and optimize machine performance. This, in turn, helps in reducing maintenance and repair costs. Vibration sensors designed for monitoring applications are used to monitor motors, critical pumps, fans, gearboxes, and compressors in the oil and gas industry.

The deterioration in the condition of machines used in **chemical plants** and their performance is usually associated with vibration, misalignment, and imbalances in moving parts; and poorly lubricated parts. Detecting these underlying problems at an early stage allows the rectification of issues before they affect different processes, thus optimizing the performance of chemical plants. Vibration monitoring equipment and tools are used in the chemicals industry to detect vibration-related issues. These tools help improve the maintenance effectiveness, increase profitability, provide safety warnings, and reduce operating costs of chemical plants. In a general chemical processing plant, vibration sensors with a minimum of 10–10,000 Hz frequencies are required for the effective monitoring of turbines, blades, and gears.

The **power industry** is facing challenges owing to the continuously increasing demand for power and the requirement to reduce distribution losses. This has led power generation companies to offer continuous power supply at low costs and monitor their processes for any unwanted changes. Hence, vibration sensors play a crucial role in the power sector. Vibration sensors help in tracking rotor imbalances, aerodynamic asymmetry, surface roughness, and overall performance, as well as carry out offline and online measurements of stress and strain in energy and power industry.

Their construction consists of a crystal of piezoelectric material to which is attached a seismic mass. When the crystal is stressed in tension or compression, it generates an electrical charge which is proportional to the acceleration level it is experiencing. Internal circuitry converts this signal into a voltage or current (4-20mA) output for data collectors or process control loops. (see figure 1)

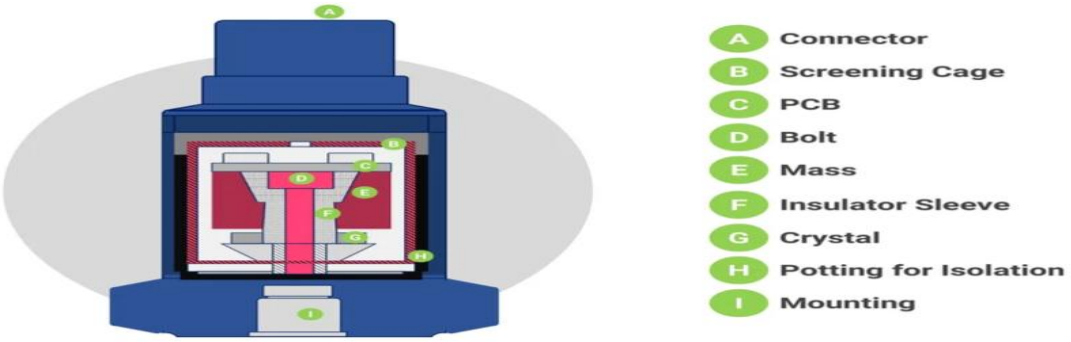
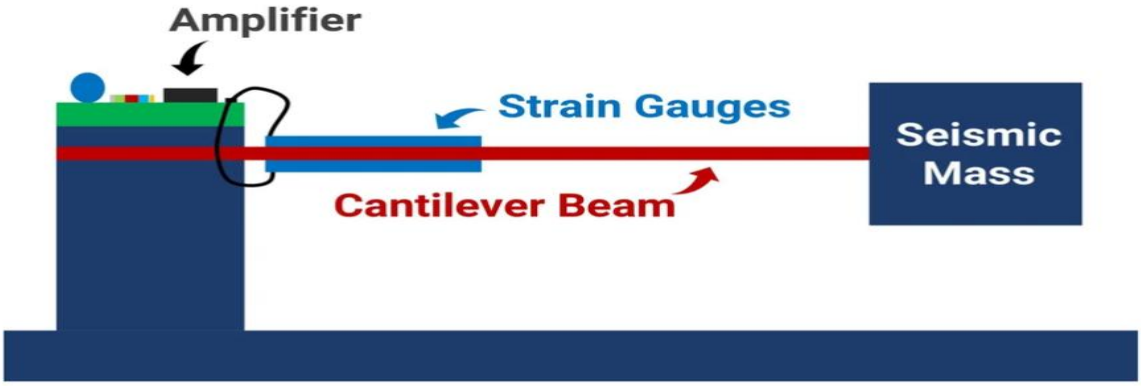


Figure 1

This robust device has no moving parts and offers long term stability and reliability. It has very wide frequency and dynamic ranges and signals can be integrated to give velocity and displacement values.

Piezoresistive accelerometers consist of a seismic mass which is attached to a cantilever beam. The beam is deflected whilst experiencing 'g' forces and this movement is converted to an electrical signal by resistance changes in a semiconductor sensing element. Internal electronic circuitry provides amplification of the signal and temperature compensation.

The frequency range of this device is lower than piezoelectric versions, but has the advantage of being able to monitor static or DC acceleration levels.



Over the last twenty-five years, different versions of vibration sensors have been designed to suit a whole new range of environmental conditions in an ever increasing need to perform consistently in differing industrial applications. These conditions have helped to create models which are primarily robust with an outer stainless-steel construction. They will operate successfully in both high and low temperatures, typically over -55 to +140 degrees C. The stainless-steel housing is welded to give protective sealing to IP68, or full liquid submersion in underwater applications. Radiation resistant versions are available, as are fully ATEX approved intrinsically safe models for operation in potentially explosive atmospheres. Cable choice for the different applications covers tough stainless-steel braiding, submersible and high temperature PTFE options. Mechanical fittings are also important for the correct fixing of the sensor to ensure correct transfer of the vibration signal and come in a choice of differing threads, adaptors, quick-fit or glue mounting studs and magnets. Custom designed

versions are also available, where a special sensor is created to exactly suit the conditions it will need to operate under.

How to use or work with a vibration sensor

Before taking a vibration measurement, you need to attach a sensor that can detect vibration behavior to the machine that is being measured. A vibration sensor either connects directly to an asset or monitors it wirelessly. Once placed, it will detect vibrations from the asset through various means, depending on the type of sensor.

When determining the correct vibration sensor for your application it is important to consider:

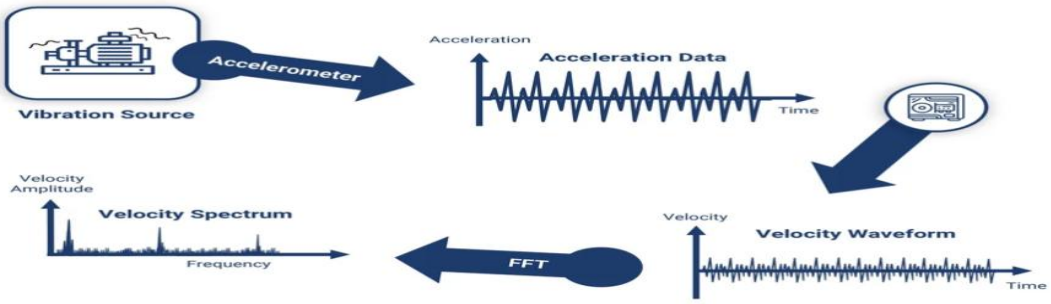
- The range and accuracy
- Environment conditions
- The shape of the measuring surface

Various types of vibration sensors are available, but a type called accelerometer is normally used as it offers advantages over other sensors. An accelerometer is a sensor that produces an electrical signal that is proportional to the acceleration of the vibrating component to which the accelerometer is attached.

What is the acceleration of a vibrating component? It is a measure of how quickly the velocity of the component is changing.

The acceleration signal produced by the accelerometer is passed on to the instrument that in turn converts the signal to a velocity signal. Depending on the user's choice, the signal can be displayed as either a velocity waveform or a velocity spectrum. A velocity spectrum is derived from a velocity waveform by means of a mathematical calculation known as the Fast Fourier Transform or FFT.

The diagram below is very simplistic explanation of how vibration data is acquired.



How is the Accelerometer Mounted?

Most machines involve rotary mechanisms. Motors, pumps, compressors, fans, belt conveyors, gearboxes, all involve rotary mechanisms and are frequently used in machines.

Most rotary mechanisms in turn have bearings that support the weight of rotating parts and bear the forces associated with rotary motion and vibration. In general, large amounts of force

are borne by bearings. It is not surprising that bearings are often the place where damage occurs and where symptoms first develop.

Vibration measurements are thus usually taken at the bearings of machines, with accelerometers mounted at or near the bearings.

Since conclusions regarding machine condition - and hence whether or not money and human safety are risked - depend on the accuracy of measurements, we must be very careful how measurements are taken. **It is important to always remember that the way in which we mount the accelerometer very much determines the accuracy of measurements.**

Why you need to monitor vibrations?

Any business that uses heavy equipment in their daily operations can benefit from monitoring vibrations. The advantages of doing so include the following:

- Understand causes of damage
- Monitor repair needs
- Keep an overall health check on equipment

Vibration sensors are an excellent way to monitor the health of key assets. They provide precise information on how individual pieces of equipment are doing, and the data they provide can help you anticipate when future maintenance needs may occur. The end result is a more streamlined maintenance process and improved machine health.

During the pandemic, critical industries such as energy and power, chemicals, food and beverages, and healthcare continue to witness an increased demand for vibration sensors to ensure safe and efficient operations of machinery. Currently, online vibration monitoring is a preferred monitoring process as it is more advanced than portable condition monitoring and provides real-time machine data to plant operators.

FLOW SENSOR

A flow sensor is an electronic device that measures the flow rate or quantity of a fluid, such as a liquid or gas, in a pipe or duct. It helps monitor and regulate flow in various applications, from industrial processes to home appliances. Flow sensor is typically connected to meters for measurement, but they can also be connected to computers and digital interfaces.

They are commonly used in HVAC systems, medical equipment, chemical plants and water treatment system. It can detect leaks, blockages, pipe burst and change in fluid concentration due to contamination and pollution.

Types of Flow Sensor

1.Differential pressure flow sensor: It measures flow rate by detecting the pressure difference between two points in a system. Here's a detailed overview:

Working Principle:

1. Pressure Taps: Two pressure taps are installed in a pipe or duct, one upstream and one downstream of a constriction or obstruction.

2. Differential Pressure Measurement: The sensor measures the pressure difference (ΔP) between the two taps.
3. Flow Calculation: The flow rate is calculated using the Bernoulli's principle, which states that the pressure drop is proportional to the square of the flow rate.

Types of Differential Pressure Flow Sensors:

1. Orifice Plates: A simple, cost-effective option using a plate with a hole to create a pressure drop.
2. Venturi Tubes: A more accurate option using a converging section to create a pressure drop.
3. Flow Nozzles: A high-accuracy option using a nozzle-shaped constriction.
4. Pitot Tubes: A type of sensor that uses a tube to measure the dynamic pressure of a fluid.

Advantages:

1. Accurate Flow Measurement: Differential pressure flow sensors provide reliable and accurate flow measurements.
2. Wide Range of Applications: Suitable for various industries and applications, including liquids, gases, and steam.
3. Cost-Effective: Can be a cost-effective solution for flow measurement.
4. Simple Installation: Relatively simple installation and maintenance.

Disadvantages:

1. Pressure Drop: The constriction or obstruction in the pipe can cause a permanent pressure drop, which can affect system performance.
2. Accuracy Dependence on Installation: Accuracy can be affected by installation conditions, such as pipe diameter, straight pipe lengths, and fluid properties.
3. Limited Turndown Ratio: May not be suitable for applications with wide flow rate ranges.

Applications:

1. Industrial Processes: Measuring flow rates of liquids, gases, and steam in various industrial processes.
2. HVAC Systems: Monitoring airflow and water flow in heating, ventilation, and air conditioning systems.
3. Water Treatment: Measuring flow rates in water treatment plants and distribution systems.
4. Oil and Gas: Monitoring flow rates in oil and gas production, processing, and transportation.

2.Thermal Flow Sensors: Thermal flow sensors measure fluid flow by detecting changes in temperature. They work on the principle that flowing fluid affects heat transfer between a heated sensor and the fluid.

Types:

1. Hot Wire Anemometers: Use a heated wire to measure flow rate.
2. Thermistor-Based Sensors: Use thermistors to measure temperature changes.

Applications:

1. Industrial processes: Measuring flow rates of gases and liquids.
2. HVAC systems: Monitoring airflow and gas flow.
3. Medical devices: Measuring respiratory flow rates.
4. Aerospace: Measuring airflow and gas flow.

Advantages:

1. High accuracy: Thermal flow sensors provide accurate measurements.
2. Fast response time: Quickly respond to changes in flow rate.
3. Low power consumption: Energy-efficient operation.

Disadvantages:

1. Fluid properties: Accuracy affected by fluid properties like temperature and composition.
2. Sensor contamination: Contamination can affect sensor accuracy.

3. Mass flow sensors: It measure the mass flow rate of a fluid (gas or liquid) in a system. They provide direct measurement of mass flow, unlike volumetric flow sensors which measure volume flow rate.

Types:

1. Coriolis Mass Flow Sensors: Use the Coriolis effect to measure mass flow rate.
2. Thermal Mass Flow Sensors: Measure mass flow rate based on heat transfer principles.

Applications:

1. Industrial processes: Measuring mass flow rates of gases and liquids.
2. Chemical processing: Accurate measurement of reactants and products.
3. Fuel injection systems: Measuring fuel flow rates in engines.
4. Medical devices: Measuring gas flow rates in respiratory devices.

Advantages:

1. Direct mass flow measurement: Accurate measurement of mass flow rate.
2. High accuracy: Mass flow sensors provide precise measurements.
3. Wide range of applications: Suitable for various industries and fluids.

Disadvantages:

1. Complexity: Some mass flow sensors can be complex and require calibration.
2. Cost: Can be more expensive than other types of flow sensors.

4. Magnetic flow sensors: It is also known as magmeters or electromagnetic flowmeters, measure the flow rate of conductive fluids. They work on the principle of electromagnetic

induction, where a magnetic field is generated and the fluid flow induces a voltage proportional to the flow rate.

How it works:

1. Magnetic field generation: A magnetic field is created around the flow tube.
2. Fluid flow: Conductive fluid flows through the tube, inducing a voltage.
3. Voltage measurement: The induced voltage is measured, proportional to flow rate.

Applications:

1. Water treatment: Measuring flow rates in water treatment plants.
2. Chemical processing: Measuring flow rates of conductive chemicals.
3. Food and beverage: Measuring flow rates of conductive liquids.
4. Wastewater treatment: Measuring flow rates in wastewater treatment plants.

Advantages:

1. High accuracy: Magnetic flow sensors provide accurate measurements.
2. Low maintenance: No moving parts, reducing maintenance needs.
3. Bi-directional flow: Can measure flow in both directions.

Disadvantages:

1. Conductive fluid requirement: Only suitable for conductive fluids.
2. Limited application: Not suitable for non-conductive fluids like gases or oils.

LEVEL SENSOR

A level sensor detects the level of a substance, such as a liquid, solid, or slurry, in a container or tank. A level sensor is a device used to monitor and measure the level of liquids, solids, or granular materials in tanks, silos, and other containers. These sensors can provide either a single point of detection (a "high" or "low" level) or continuous measurement of the substance's level as it fluctuates. There are various types of level sensors, including:

1. **Float level sensors:** (Use a floating element to detect level changes) These use a buoyant object (a float) that moves with the level of the liquid.

- **How they work**: A magnet within the float activates a hermetically sealed switch in the sensor stem, or the float's movement directly triggers a mechanical switch.
- **Best for**: Water, oil, and fuel tanks in simple applications.
- **Considerations**: Can be unreliable with high-viscosity liquids or in turbulent conditions.
- **Example product**: The RS PRO Float Switch (Model 1919464) uses a float to provide a high or low level indication.

2**. Ultrasonic level sensors**: (Use sound waves to measure level) These non-contact sensors use high-frequency sound waves to measure the distance to the material's surface.

- **How they work**: A transducer emits a sound wave and measures the time it takes for the echo to return after bouncing off the surface. This time is used to calculate the distance to the surface.
- **Best for**: Non-contact measurement of viscous liquids and bulk solids, as well as water treatment facilities.
- **Considerations**: Accuracy can be affected by foam, vapor, turbulence, and rapid temperature changes.
- **Example product**: The RS PRO Ultrasonic / Sonar Level Sensor (Model 2565734) provides a non-contact solution for liquid level sensing.

3. **Capacitance level sensors**: (Measure changes in capacitance to detect level) These versatile sensors measure the change in electrical capacitance caused by the presence of a substance.

- **How they work**: A probe inserted into the tank forms a capacitor with the tank wall. As the material level changes, the capacitance changes, which is then converted into a level measurement.
- **Best for**: Both conductive and non-conductive liquids, slurries, and solids. They work well in high-temperature and high-pressure environments.
- **Considerations**: May require calibration for different materials and can be affected by material buildup on the probe.
- **Example product**: The Taidacent Non-Contact Liquid Level Sensor can be attached externally to a non-metallic container to detect liquid levels.

4. **Radar level sensors:** (Use radar waves to measure level) Using microwave pulses, radar sensors measure the time-of-flight of radio waves to the material surface.

- **How they work**: An antenna on the sensor emits microwaves that reflect off the surface and return. The time interval is directly proportional to the material's level.
- **Best for**: Harsh environments with extreme temperature, pressure, vapor, or dust, as microwaves can penetrate these conditions more effectively than sound waves.
- **Considerations**: Generally more expensive than ultrasonic sensors.
- **Example product**: Radar-based options are available from manufacturers like **RS PRO** and are highly accurate.

5. **Pressure level sensors**: Measure pressure changes to detect level.

Applications:

1. **Industrial processes:** Monitoring liquid levels in tanks and vessels.
2. **Water treatment:** Measuring water levels in treatment plants.
3. **Chemical processing:** Monitoring chemical levels in tanks.
4. **Food and beverage:** Measuring liquid levels in tanks and vessels.

5. **Oil and gas**: Monitoring crude oil levels in storage tanks.
6. **Water and wastewater treatment**: Regulating water levels in reservoirs, settling basins, and storage tanks.
7. **Food and beverage**: Ensuring precise filling levels in vats and packaging.
8. **Chemical and pharmaceutical**: Monitoring chemical levels in storage and processing vessels to prevent spills.
9. **Automotive**: Measuring levels for engine oil, coolant, and fuel.
10. **Agriculture**: Monitoring water and fertilizer levels in storage tanks for irrigation.

Benefits:

1. Accurate level measurement: Level sensors provide precise measurements.
2. Real-time monitoring: Enable real-time monitoring and control.
3. Automation: Can be integrated with automation systems for efficient operation.

HUMIDITY SENSOR

A humidity sensor, or hygrometer, is an electronic device that measures and reports the amount of water vapor in the air. These devices are used across multiple industries to maintain specific environmental conditions, prevent equipment damage, and improve comfort.

Types of humidity sensors:

1. **Capacitive humidity sensors:** These are the most common and versatile type of humidity sensor.

   How they work: A capacitive sensor measures relative humidity (RH) by using a thin polymer film, or dielectric material, placed between two electrodes. The film absorbs moisture from the air, which changes the electrical capacity (capacitance) between the two electrode plates.

   Key characteristics:

   - Typically highly accurate and stable for long-term use.
   - Can measure the full range of relative humidity, from 0% to 100%.
   - Used in a wide range of applications, including HVAC systems, weather stations, and consumer electronics.

2. **Resistive humidity sensors:** These are a simple and low-cost alternative to capacitive sensors.

   How they work: A resistive sensor uses a conductive, moisture-absorbing material, such as a polymer or salt, placed between two electrodes. As the material absorbs moisture, its electrical resistance decreases. The sensor measures this resistance to determine the relative humidity.

   Key characteristics:

   - Compact and inexpensive, making them suitable for residential and industrial applications that do not require extremely high precision.

- Can be sensitive to contaminants, which may affect accuracy over time.

3. **Thermal conductivity humidity sensors:** This type of sensor is designed to measure absolute humidity (AH).

   How they work: A thermal sensor uses two thermistors (resistors that change resistance with temperature). One thermistor is sealed in a chamber of dry nitrogen, while the other is exposed to the ambient air. The sensor measures the difference in thermal conductivity between the two thermistors, as humid air conducts heat better than dry air.

   Key characteristics:

   - Durable and able to function in high temperatures and corrosive environments.
   - Can provide higher resolution at higher temperatures compared to other types.

Applications:

Humidity sensors are used to monitor and control moisture levels in a variety of environments.

- **Residential and commercial**: Integrated into HVAC systems to optimize energy efficiency and indoor air quality by preventing mold growth and ensuring comfort. They are also found in smart home devices like thermostats and weather stations.
- **Agriculture**: Used in greenhouses to monitor and control humidity levels for optimal crop growth and to prevent fungal diseases. Soil moisture sensors can also use capacitive technology.
- **Industrial manufacturing**: In settings like pharmaceuticals, food processing, and electronics manufacturing, precise humidity control is critical for product quality. Sensors can prevent issues like moisture-related corrosion or spoilage.
- **Data centers**: Control humidity levels in server rooms to prevent equipment damage from electrostatic discharge or corrosion caused by excess moisture.
- **Automotive**: Used in vehicles for climate control systems and to prevent windshields from fogging.
- **Healthcare**: Maintain proper humidity levels for storing medical supplies and medications. They are also used in medical equipment such as incubators and ventilators.

PRESSURE SENSOR

A pressure sensor is an electronic device that measures the force exerted by a gas or liquid and converts it into an electrical signal. Pressure sensors are also commonly referred to as pressure transducers or transmitters, though these terms can refer to specific types of sensors with different output signals. They are used to monitor and control pressure in many applications, from medical devices to industrial automation.

Types of pressure measurements

Pressure sensors are defined by the type of pressure they measure relative to a specific reference point.

- **Absolute pressure sensors:** Measure pressure relative to a perfect vacuum, where pressure is zero. The reading is not affected by changes in atmospheric pressure.

- **Gauge pressure sensors:** Measure pressure relative to the local atmospheric pressure. The reading can change with weather or altitude. Most everyday applications, like checking tire pressure, use gauge pressure.
- **Differential pressure sensors:** Measure the difference between two separate pressure points in a system. They are often used to measure the flow of liquid or gas through pipes.
- **Sealed pressure sensors:** Measure pressure relative to a fixed pressure sealed inside the sensor, such as atmospheric pressure at sea level.
- **Vacuum pressure sensors:** Measure pressure that is lower than the surrounding atmospheric pressure.

Common pressure-sensing technologies

The mechanical displacement of a diaphragm due to pressure is converted into an electrical signal using several sensing technologies.

- **Piezoresistive:** As a diaphragm deforms under pressure, strain gauges attached to it change their electrical resistance, which is then measured. This method is rugged, stable, and commonly used in industrial and automotive systems.
- **Capacitive:** A diaphragm and a fixed plate form a capacitor. As the diaphragm moves with pressure, the distance between the plates changes, altering the capacitance. These sensors offer high sensitivity and low power consumption, making them ideal for low-pressure applications.
- **Piezoelectric:** Certain materials, like quartz or ceramics, generate an electrical charge when subjected to mechanical stress. This makes them suitable for measuring dynamic, fast-changing pressures, such as engine combustion.
- **MEMS (Micro-Electro-Mechanical System):** These are small, micro-fabricated sensors on a silicon chip. They can be based on piezoresistive or capacitive principles and are used in consumer electronics and medical devices.

TEMPERATURE SENSOR

A temperature sensor is an electronic device that measures the temperature of its surroundings and converts the data into an electrical signal. The two main categories are contact and non-contact sensors. Contact sensors measure the temperature of an object through direct physical contact, while non-contact sensors measure thermal radiation to determine the temperature from a safe distance.

How to choose the right sensor

1. Accuracy 2, Cost 3, Temperature range 4. Response time
2. Environmental conditions

Sensor Type	Operating Principle	Common Uses	Key Characteristics
Thermocouples	Two dissimilar metals are joined at one end. A change in temperature at the joined end creates a voltage across the open ends (the Seebeck effect).	Industrial furnaces, automotive diagnostics, and gas appliance pilot lights.	Inexpensive, durable, wide temperature range (-200 °C to 1750 °C), but lower accuracy.
Resistance Temperature Detectors (RTDs)	The electrical resistance of a metal, typically platinum, changes linearly with temperature.	Precision applications in the food industry, medical devices, and industrial processes.	Highly accurate, stable, and repeatable measurements, but more expensive and slower response times than other types.
Thermistors	The resistance of a semiconductor material changes significantly with temperature.	Household appliances (refrigerators, air conditioners) and digital thermometers.	Cost-effective and highly sensitive to small temperature changes. Can be either NTC (resistance decreases with temperature) or PTC (resistance increases).
Infrared (IR) sensors	Detect an object's temperature by measuring its emitted infrared radiation.	Remote temperature monitoring in high-voltage or high-temperature environments.	Non-contact measurement is ideal for hazardous conditions, but accuracy depends on cost.
Semiconductor sensors	An integrated circuit (IC) uses the temperature-sensitive voltage characteristics of diodes to provide an output voltage or digital reading.	Computer processors, consumer electronics, and portable devices.	Small size, linear output, and low cost, with a limited temperature range and slower response time than other sensor types.

Applications:

1. Industrial Control: Temperature sensors are used in industrial control applications, such as monitoring temperature in manufacturing processes, chemical processing, and oil refining.

2. HVAC Systems: Temperature sensors are used in heating, ventilation, and air conditioning systems to regulate temperature and maintain a comfortable environment.

3. Automotive: Temperature sensors are used in vehicles to monitor engine temperature, cabin temperature, and other critical temperatures.

4. Medical Devices: Temperature sensors are used in medical devices, such as patient monitors, incubators, and medical imaging equipment.

5. Aerospace: Temperature sensors are used in aerospace applications, such as monitoring temperature in aircraft engines and other critical systems.

Benefits:

1. Improved Safety: Temperature sensors can help prevent overheating and other temperature-related issues that can lead to equipment damage or safety hazards.

2. Increased Efficiency: Temperature sensors can help optimize industrial processes and HVAC systems, leading to increased efficiency and reduced energy consumption.

3. Enhanced Product Quality: Temperature sensors can help ensure that products are manufactured within a specific temperature range, which can affect product quality and reliability.

Selection Criteria:

1. Temperature Range: The temperature range of the sensor should match the application requirements.

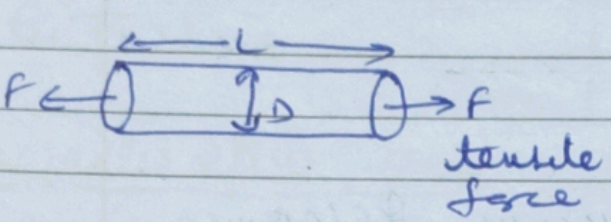
2. Accuracy: The accuracy of the sensor should meet the application requirements.

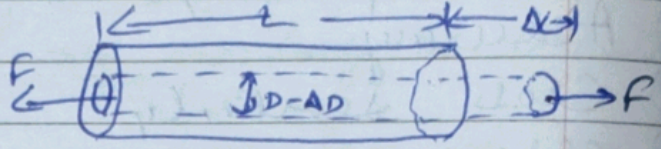
3. Response Time: The response time of the sensor should be suitable for the application.

4. Environmental Factors: The sensor should be able to withstand environmental factors such as humidity, vibration, and corrosion.

Theory of Strain Gauges

The change in value of R by straining the gauge may explain partly normal behaviour of elastic material





if strip of elastic material is subjected to tension or positively strained, its longitudinal dimension will rise while there will be reduction in lateral dimension.

So, when gauge is subjected to the strain, its length rises and its area of cross section falls.

Since

$$R \text{ of conductor} \propto \text{length} \propto \frac{1}{\text{area of cross section}}$$

Then R rises of gauge with the strain

Let consider strain gauge made of circular wire. The wire has dimension

length = L, area = A, diameter = D

before strained

the material of wire has resistivity ρ

$\therefore$ R of unstrained gauge $R = \frac{\rho L}{A}$ — (1)

Then tensile strength S be applied to wire. This produces the strain causing length to rise and area to fall. So, when wire is strained, then all change in dimension

Let ΔL = Change in length
ΔA = " " Area
ΔD = " " diameter
ΔR = " " Resistance

In order to find out ΔR depend on material physical Quantity, the eq of R is differentiated wrt stress S, Then,

$$\frac{dR}{dS} = \frac{\rho}{A}\frac{\partial L}{\partial S} - \frac{\rho L}{A^2}\frac{\partial A}{\partial S} + \frac{L}{A}\frac{\partial \rho}{\partial S} \quad \text{---(2)}$$

from dividing eq (2)/eq (1)

$$\frac{1}{R}\frac{dR}{dS} = \frac{1}{L}\frac{\partial L}{\partial S} - \frac{1}{A}\frac{\partial A}{\partial S} + \frac{1}{\rho}\frac{\partial \rho}{\partial S} \quad \text{---(3)}$$

from (3), Per unit change in R is due to:

⊕ Per unit change in length $= \Delta L/L$
⊕ " " " " Area $= \frac{\Delta A}{A}$
⊕ " " " " resistivity $= \frac{\Delta \rho}{\rho}$

we know

$$A = \frac{\pi}{4}D^2$$

$$\therefore \frac{\partial A}{\partial S} = 2 \cdot \frac{\pi}{4} \cdot D\frac{\partial D}{\partial S} \quad \text{---(4)}$$

or $$\frac{1}{A}\frac{\partial A}{\partial S} = \frac{(2\pi/4)D}{\pi/4 D^2}\frac{\partial D}{\partial S} = \frac{2}{D}\frac{\partial D}{\partial S} \quad \text{---(5)}$$

Then eq (3) will be

$$\frac{1}{R}\frac{dR}{dS} = \frac{1}{L}\frac{\partial L}{\partial S} - \frac{2}{D}\frac{\partial D}{\partial S} + \frac{1}{\rho}\frac{\partial \rho}{\partial S} \quad \text{---(6)}$$

Now

Poisson ratio $\vartheta = \frac{\text{lateral strain}}{\text{longitudinal strain}} = \frac{-\partial D/D}{\partial L/L}$

So, $$\frac{\partial D}{D} = -\vartheta \times \frac{\partial L}{L}$$

So, eqn (1) becomes

$$\frac{1}{R}\frac{dR}{ds} = \frac{1}{L}\frac{\partial L}{\partial s} + \nu\frac{2}{L}\frac{\partial L}{\partial s} + \frac{1}{\rho}\frac{\partial \rho}{\partial s} \quad \text{— (2)}$$

for small variation eqn (2) becomes -

$$\frac{\Delta R}{R} = \frac{\Delta L}{L} + 2\nu\frac{\Delta L}{L} + \frac{\Delta \rho}{\rho}$$

Gauge factor is defined as ratio of per unit change in resistance to per unit change in length.

$$\text{Gauge factor } G_f = \frac{\Delta R/R}{\Delta L/L}$$

$$\frac{\Delta R}{R} = G_f \cdot \frac{\Delta L}{L}$$

$$\boxed{\frac{\Delta R}{R} = G_f \times \varepsilon}$$

where

$$\varepsilon = \text{Strain} = \frac{\Delta L}{L}$$

So, Gauge factor G_f can be written as:

$$G_f = 1 + 2\nu + \frac{\Delta\rho/\rho}{\varepsilon}$$

$$G_f = 1 + 2\nu + \frac{\Delta\rho/\rho}{\varepsilon}$$

- 1 → Resistance change due to change in length
- 2ν → Resistance change due to change in area
- $\frac{\Delta\rho/\rho}{\varepsilon}$ → Resistance change due to piezo-resistive effect

$$G_f = \frac{\Delta R/R}{\Delta L/L} = 1 + 2\nu + \frac{\Delta \rho/\rho}{\Delta L/L}$$

Strain is generally expressed in microstrain
$= 1\ \mu m/m$

So, if change in value of resistivity of material when strained is neglected, then

Gauge factor $\boxed{G_f = 1 + 2\nu}$ — (8)

Poission ratio for all metal → 0 - 0.5
Common value of poission ratio for wire is 0.3